Chapter:-7

Find the number and symmetry species of the Raman and IR active vibration of BF_3 (D_{3h})

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	12	0	-2	4	-2	2

According to the character table of D_{3h} point group.

No. of $A_1' = \frac{1}{12}[(12\times1\times1)+0(-2\times1\times3)+(4\times1\times1)$
$+(-2\times1\times2)+(2\times1\times3)]$
$= 1$

Reduces to $A_1' + A_2' + 3E' + 2A_2'' + E''$

According to character table of D_{3h}.

~~Rotation~~

Symmetry of all molecular motions :

$A_1' + A_2' + 3E' + 2A_2'' + E''$

Rotations : $A_2'' + E''$

Translations : $A_2'' + E'$

vibrations : $A_1' + 2E' + A_2''$

~~Raman Active :~~

IR active : $2E' + A_2''$

Raman active : $A_1' + 2E'$.

IR active → x^2+y^2, z^2 ... → x, y, z

Raman active → ~~Raman~~ multiple ways

↳ xy, yz, xz.

Translational → x, y, z.

Rotational → Rx, Ry, Rz.

for H_2O molecule

C_{2v}	E	C_2	$\sigma_v(xy)$	$\sigma_v'(yz)$
Γ	9	−1	3	1

From the character table of C_{2v} we get,

$\Gamma = 3A_1 + A_2 + 3B_1 + 2B_2$.

So, symmetry of all molecular motion-

~~$3A_1$~~ $3A_1 + A_2 + 3B_1 + 2B_2$

Symmetry of translations :

$A_1 + B_1 + B_2$

Symmetry of Rotational :

$A_2 + B_1 + B_2$

∴ symmetry of vibrations $2A_1 + B_1$.

$2A_1 + B_1$..

IR active = $A_1 + B_1$

Raman active = B_1. A_1 B_1

for IR active → x, y, z

Raman → xy, yz, zx, x^2-y^2, z^2.

xef4.

D_{4h}	E	$2C_4$	$C_2(C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	15	1	−1	−3	−1	−3	−1	−5	3	1

from the Dnh character table we get,

$\Gamma = A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$

so, symmetry of all molecular motion:

$A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$.

Symmetry of translations : $A_{2u} + E_u$

Symmetry of rotations :

$A_{2g} + E_g$

Symmetry of vibrations -

$A_{1g} + B_{1g} + B_{2g} + A_{2u} + B_{2u} + 2E_u$.

IR active → $A_{2u} + E_u$

Raman active → $A_{1g} + B_{1g} + B_{2g}$

fore NH_3

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	12	0	2

From the character table of NH_3 we get,

No. of $A_1 = \frac{1}{6}[(12\times1\times1)+0+(2\times1\times3)]$

$= 3$

No. of $A_2 = \frac{1}{6}[(12\times1\times1)+0+\{2\times(-1)\times3\}]$

$= 1$

No. of $E = \frac{1}{6}[(12\times2\times1)+0+(2\times0\times3)]$

$= 4$

So, we get, $\Gamma = 3A_1 + A_2 + 4E$.

So, symmetry of all molecular motion,

$3A_1 + A_2 + 4E$

Translation : A_1 E

Rotational : A_2 $+E$

Vibrational : $2A_1 + 2E$

IR active $\longrightarrow$ $2A_1 + 2E$

Raman active $\longrightarrow$ $2A_1 + 2E$

(5.) H_2O

C_{2v}	E
Γ	2